

PYTHON FOR BEGINNERS · DAY 2

Variables & Data Types

Storing information, input & operators

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```
name = "Homaira"
age = 20
gpa = 3.8

print(name, age, gpa)
```

TODAY'S GOALS

By the end of today, you can…

- Create and update **variables**
- Use **int**, **float**, **str**, **bool**
- Get input with **input()**
- Follow the naming rules
- Do maths with operators
- Convert with **int()** and **float()**

REVIEW OF LAST CLASS

What did we learn?

Python basics

What Python is and where it’s used

print()

Shows output on the screen

Comments

Notes that start with #

Running code

Top to bottom, line by line

```
print("Hello Python")
```

WHY DO WE NEED VARIABLES?

Store it once, use it anywhere

Without variables 😩

```
print("Homaira")
print(20)
print("Computer Science")
```

Hard to manage, hard to change, repeated data

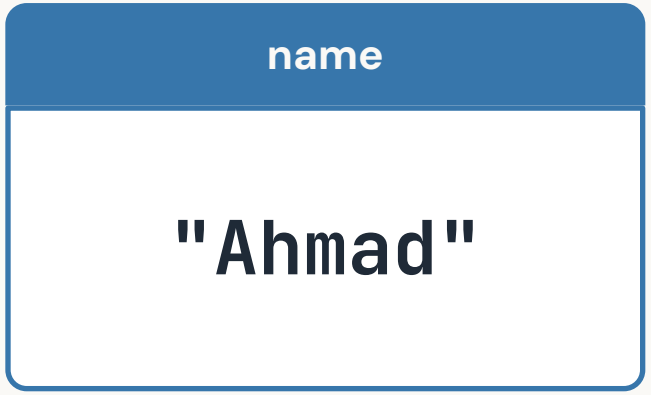
With variables 😎

```
name = "Homaira"
age = 20
major = "Computer Science"
```

Stored cleanly — easy to manage and update

WHAT IS A VARIABLE?

A labelled box that stores a value



The **label** is the variable's name

What's **inside** is its value

```
name = "Ahmad"
```

CREATING VARIABLES

variable_name = value

Code

```
student_name = "Sara"
student_age = 19
student_gpa = 3.5

print(student_name)
print(student_age)
print(student_gpa)
```

Output

```
Sara
19
3.5
```

= means “store this value”, **not** “is equal to”.

NAMING RULES

Good names, allowed names

Rule	Wrong	Correct
Can’t start with a number	1name	name1
No spaces — use _	student name	student_name
Case-sensitive	Name = name	Name ≠ name
Use clear names	x = 20	student_age = 20

Python style: all lowercase with _ between words (snake_case).

VARIABLES CAN CHANGE

Multiple assignment & updates

Many at once

```
x, y, z = 10, 20, 30
print(x, y, z)
```

Same as three separate lines.

Change the value

```
score = 50
score = 80      # replaced
score += 10     # now 90
```

+= is short for score = score + 10

DATA TYPES

What kind of information is it?

`int`

Whole number

`10`

`float`

Decimal

`3.5`

`str`

Text (string)

`"Python"`

`bool`

True or False

`True`

```
type(20) → int
```

```
type("20") → str ⚠️
```

ARITHMETIC OPERATORS

Maths with a = 7 and b = 2

+	-	*	/	//	%	**
add	subtract	multiply	divide	whole part	remainder	power
9	5	14	3.5	3	1	49

% is very useful: if **number % 2** is 0, the number is **even**. We use this on Day 3!

INPUT()

Getting information from the user

Code

```
name = input("What is your name? ")
print("Hello", name)
```

Output

```
What is your name? Ali
Hello Ali
```

The program **waits** until the user types and presses **Enter**. The text in the brackets is the question (the **prompt**).

⚠️ VERY IMPORTANT

input() always gives text

Even if the user types a number, it's stored as a **string**.

```
age = input("Age: ")
# user types 20

print(type(age))
# <class 'str'>

print(age + 5)
# ❌ TypeError
```

TYPE CONVERSION

Turn text into a number

`int("20")`
→ 20 (whole number)

`float("3.5")`
→ 3.5 (decimal)

`str(20)`
→ "20" (text)

```
age = int(input("Age: "))
print("Next year:", age + 1)
```

Read it **inside-out**: input() gets "20" → int() makes it 20 → stored in age.

TEXT + VARIABLES

Three ways to combine them

1. Commas ✅

```
print("Age:", age)
```

Easiest. Adds spaces for you.

2. Plus +

```
print("Age: " + str(age))
```

Only joins strings — numbers need str().

3. f-strings ⭐

```
print(f"Age: {age}")
```

The modern way: variables go in { }.

🧠 QUICK CHECK

Predict before you run!

Q1. What is printed?

```
score = 50
score = score + 25
print(score)
```

Q2. Which type?

```
type("3.14")
```

Q3. What is printed?

```
print(17 % 5)
```

💻 LET'S CODE TOGETHER

Practice time!

Store your name, age and major, mix all 4 data types, and print everything. Then open **exercises.ipynb**.

⭐ About me variables

⭐ ⭐ Sharing sweets with // and %

⭐ ⭐ ⭐ Shopping receipt 🧾

Next lesson: Making decisions with if 🧠